

Rupjyoti Barman

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Summary

.NET Backend Developer with **3 years** of experience designing and delivering **hardware diagnostic APIs** and **system-level platforms** for **HP PC Hardware Diagnostics** — HP's official diagnostic application. Proven expertise in **C#, ASP.NET Core**, and **managed-unmanaged interoperability** (P/Invoke, Native C++ DLL integration), with hands-on experience in **decoupled NuGet-based architectures**, **WMI-based real-time monitoring**, and **Windows System APIs** (WUA, HPCMSL, DeviceIoControl). Experienced in **Clean Architecture**, **Factory pattern**, **Entity Framework Core**, and **Azure Pipelines**, with a track record of eliminating third-party release dependencies, reducing diagnostic development cycles, and delivering production-grade features to **cross-functional global teams** across India, Taiwan, and the US.

Technical Skills

- **Languages:** C#, C++, SQL, JavaScript, Python
- **Frameworks & Libraries:** ASP.NET Core (Web API, MVC), Entity Framework Core, WPF, React, Bootstrap, jQuery, LINQ
- **Windows System APIs:** WMI, WQL, ManagementEventWatcher, WUA API, HPCMSL, DeviceIoControl, P/Invoke, Win32 APIs
- **Architecture & Design:** Clean Architecture, Layered Architecture, Microservices, SOLID Principles, OOP, Factory Pattern, Dependency Injection, Repository Pattern
- **Testing:** MSTest, Moq, WPF-based API Testing
- **Database:** SQL Server, PostgreSQL, Entity Framework Core (Code-First, Migrations)
- **DevOps & Tools:** Azure Pipelines, Git, Bitbucket, NuGet, Visual Studio, Postman, Jira, Confluence
- **Security & Logging:** JWT Authentication, Role-Based Access Control (RBAC), Serilog

Experience

Software Developer, BoldTek India Pvt. Ltd. – Guwahati

June 2023 – Present

Client: Hewlett-Packard (HP)

- Developed 2 production features in an **ASP.NET Core MVC** application (Controller, Service, Repository architecture with **Entity Framework Core Code-First** migrations and **SQL Server**) with **JavaScript** and **Bootstrap** frontend; implemented an **ODM self-service dashboard** with role-based access control (Viewer, ODM, Admin) enabling **INI configuration** leveraging, verification submission and real-time approval status tracking, backed by **INI key-value validation logic** across **1000+ HP NPI configuration files**; built a **Release Notes Dashboard** with structured data entry against release versions, replacing **manual Excel-based tracking** and enabling **HP Factory Diagnostic team** to query release-specific INI changes directly from **SQL Server** for **HP NPI factory diagnostic systems**.
- Led the India team in migrating from a **shared RPC-based Windows Service architecture** to a **decoupled 3-layer NuGet package (Business Logic, C# Interop Wrapper via P/Invoke, Native C++ DLL)** with **Factory pattern**-based dependency injection, eliminating shared service dependency across multiple HP teams; achieved **100% elimination of third-party release delays** (previously **10–15 days**) and reduced **diagnostic development cycle from 4 to 3 sprints (40 to 30 working days, a 25% reduction)** by removing 2 redundant architectural layers (Service layer and Service Manager routing layer), while resolving critical service instability caused by cross-team unhandled exceptions and **sleep/hibernate-induced service failures**, consumed by **HP PC Hardware Diagnostics**.
- Migrated **6 hardware diagnostic modules** (Storage: **DRV Read/Verify**, **SMART Health Monitor**, **SMART Data Collection**; Network: **NIC IRQ Verification**, **ROM/MAC Validation**; Audio: **Audio Playback Test**) from legacy **RPC-based Windows Service** onto the **decoupled 3-layer NuGet architecture** and additionally designed **2 new diagnostic modules from scratch** — **Driver Update Detection (Windows Update Agent API)** and **BIOS Update Check (HP CMSL, commercial systems)** — consolidating all into a **centralized diagnostic backend** consumed by **HP PC Hardware Diagnostics**, architected to support multiple HP client applications with zero external service dependency.
- Replaced a service-based device discovery architecture (which required a **2–3 minute service restart** to reflect any hardware change) by engineering a **real-time WMI monitoring module (ManagementEventWatcher + WQL on Win32_PnPEntity)** packaged into the shared **Common** library; reduced device change detection latency **from 2–3 minutes to under 4 seconds (~97% reduction)**, with the module adopted across **multiple hardware diagnostic APIs** and consumed by **HP PC Hardware Diagnostics**, HP's official diagnostic application, eliminating the service restart dependency entirely.

- Managed **Azure Pipelines** to trigger **NuGet package builds**, publishing **Business Logic** package (with required dependency NuGets) integrated into the **C# Wrapper** layer to produce the **final consolidated NuGet package** (bundling all layers including Native **C++ DLL**) for testing and client consumption.
- Validated all migrated and newly developed diagnostic modules using a dedicated **WPF-based API testing application** for end-to-end UI-level verification and **MSTest** unit test projects across **Business Logic** and **C# Wrapper** layers, leveraging **Moq** for dependency mocking to ensure diagnostic reliability prior to client integration.
- Collaborated with cross-functional global teams (**India, Taiwan, US**) to deliver features and resolve production issues.

Education

B.Tech in Computer Science & Engineering - CGPA: 8.01/10

May 2023

North Eastern Regional Institute of Science and Technology (NERIST), Arunachal Pradesh